Green Earth Wire?

Main Power Input

J1 200241-1112 Mega-Fit

F1 3557-10 40A

D1 SMAJ51CA

50A Main Power Relay

K1 J115F11CH5VDC561.5U Coil 5V Coil 300mA

R40 180 — Must be high wattage res

D4 SMD26PL-TP

IC1 SI2318CDS-T1-GE3

R6 10k

D10 LTST-C150KRKT

R31 10k

Large DC-DC Converters (With Heatspreader)

F6 0678L9250-02

PS1 I7C4W008A120V-0C3-R — Computer Power Supply

C1 C2 C3 C5 C7 180uF/10uF 63V

C9 C11 C15 100uF 63V

C17 0.1uF 63V

TP4 TestPoint

+15V

R1 4.64k

R3 1k

C13 0.1uF 63V

IC4 SI2318CDS-T1-GE3

R9 10k

F7 0678L9250-02

PS2 I7C4W008A120V-0C3-R — POE Switch Power Supply

C4 C25 C26 C6 C8 180uF/10uF 63V

C10 C12 C16 100uF 63V

C18 0.1uF 63V

TP5 TestPoint

+48V

R2 28

R4 1k

C14 0.1uF 63V

IC5 SI2318CDS-T1-GE3

R10 10k

Power Outputs

F4 3557-10 — 7.5A Blade Fuses — Computer Power

J14 43650-0228 Micro-Fit

J4 PJ-063BH

D13 LTST-C150KRKT

R34 10k

F5 3557-10 — 5A — POE Switch Power

D14

J3 43650-0415 Micro-Fit

R35 10k

F9 3557-10 — TBD — 48V Aux PWR Out

KeySwitch Input

J10 43650-0228 Micro-Fit

TP1 KeySwitch IN

R5 28

D7 SMD26PL-TP

D2 SMAJ51CA

C19 0.1uF

R41 10k

D18

Wakeup Bootstrap Ckt

KEY SW IN

R11 10k

D8 SMD26PL-TP

Switched PWR IN PROT

MOSFET Driver

Q5 SQJ182EP-T1_GE3

R50 0

R51 100k

R52 0

C45 0.1uF

U2 AP74502HQTA8-7

C44 0.1uF

R12 10k

Buck Regulator 5.3V 600mA

U10 LV2862YDDC

C41 10uF 63V

R47 100k

C49 0.1uF

C43 0.1uF

L2 SRR1208-470YL 2A 47uH

D11 SK220A 200V 2A

C42 10uF 63V

R48 59k

R49 10k

Vout = 0.765V (1+ (R1 / R2))

D20 PPS1545-3G — 5V Logic PWR

D21 PPS1545-3G — 5V Relay PWR

Jetson Header

J7 XG4A-4035

Auxiallary Jetson Header

J11 PPPC202LFBN-RC

Rpi Pico

U1 SC0915

SW1 SW_Push — Reset Button

SW2 SW_Push — Aux SW

JP3 Jmp_2_Bridged — Jet_UART1_Rx

JP4 Jmp_2_Bridged — Jet_UART1_Tx

D9 SMD26PL-TP — 5V Logic PWR

C30 180uF 63V

C46 C47 10uF 63V

Added new functions
GP14 LED_RED
GP3 LED_YELLOW
GP4 LED_GREEN

D15 D16 D17

R36 R37 R38 1k

Temp Sense ICs

U6 U7 MCP9700T-ETT

C39 C40 0.1uF

TP9 TP11 TestPoint

ADC MUX

U5 74HC4051

C38 0.1uF

TP14 TestPoint

JP7 JP5 JP6

TP7 TestPoint

ADC Voltage Divider

R21 R22 220k/10k — KeySw_Vin_MES_ADC — TP6 TestPoint

R23 R24 220k/10k — POE_SW_Vout MES_ADC — TP13 TestPoint

R27 R28 220k/10k — COMP_Vout MES_ADC — TP8 TestPoint

External Status Light Power Supply 24V

U8 LV2862YDDC

C34 C35 10uF 63V

R46 100k

C48 0.1uF

C33 0.1uF

L1 SRR1208-470YL 2A 47uH

D6 SK220A 200V 2A

C36 C37 1uF

R44 300k

R45 10k

Vout = 0.765V (1+ (R1 / R2))

External Status Light

External LED status light
DC24V input
0.02A Red | 0.02A Yellow | 0.01A Green
4 wires
#1 brown wire = Red LED
#2 white wire = Yellow LED
#3 blue wire = Green LED
#4 black wire = Common line DC24V

J13 43650-0415 Micro-Fit — BROWN_WIRE / WHITE_WIRE / BLUE_WIRE / BLACK_WIRE

F10 SF-1206SA250M-2

Q4 Q3 Q2 2N7002

R43 R39 R32 10k

Mounting Holes

H1 H2 H3 H4 MountingHole_Pad

FID1 FID2 FID3 Fiducial

--Change log from rev 0.3--
Removed light bank relay A (F2, K2, D5, IC2, R7, D11, R32)
Removed light bank relay B (F3, K3, D6, IC3, R8, D12, R33)
Removed connector J2, J8, J9
Changed J10 to a 2 pin Molex Micro-Fit
Changed J3 to a 4 pin Molex Micro-Fit
Changed J13 to a 4 pin Molex Micro-Fit
Remove External relay circuit (IC6)
Added external status light 3 colors circuit
Changed U4 from R1560S501B-E2-FE to MCP1793T-5002H/OT because no stock
Added gate driver circuit to replace sticking relay
Changed LT1910ES8 to IRS20752L because of no stock
Added LED for keyswitch
Removed J12 Pico breakout
Removed J6 Jetson breakout

--Change log from rev 0.4--
Changed MOSFET driver to AP74502
Removed 5V LDO because the AP74502 EN is 80V tolerant

Mark Funderburk
Precesion Sustainable Agriculture
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File: System Management Board.kicad_sch

Title: System Management Board

Size: A2	Date: 2026-10-12	**Rev: V0.5**
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